CM4 eMMC computer, display, USB audio and bridge carrier

Rev F integrated keyboard computer. See validation reports.

CM4 GPIO and power

File: cm4_0.kicad_sch

CM4 USB and DSI

File: cm4_1.kicad_sch

Carrier power and display protection

File: power.kicad_sch

DSI1 four-lane display socket

File: display.kicad_sch

USB peripheral audio and recovery

File: usb.kicad_sch

Stereo I2S output

File: audio.kicad_sch

Isolated CM4 to always-on bridge

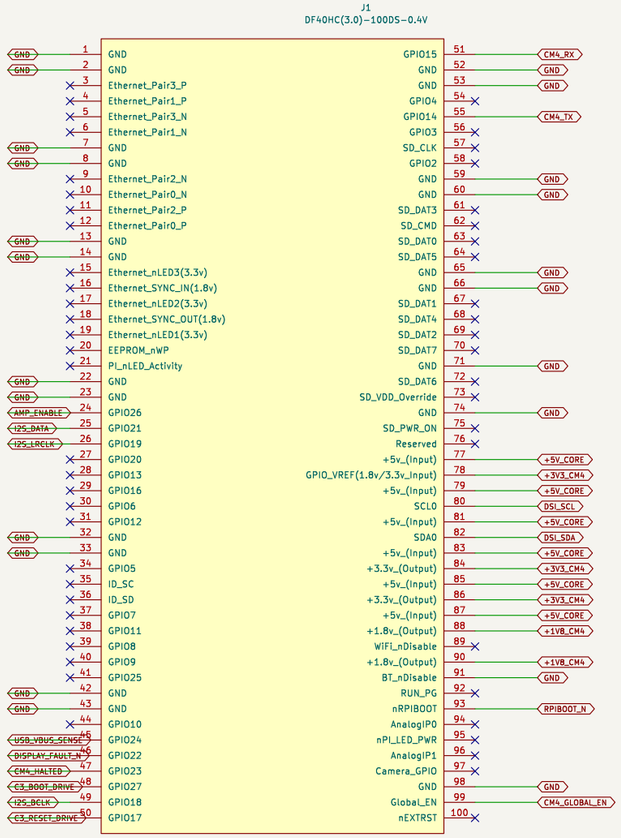
File: bridge.kicad_sch

Sheet: /
File: cm4-carrier.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 8/8

CM4 with eMMC (not Lite): two Hirose 100-position sockets, 33.92mm centre spacing.
Pin mapping transcribed from the official CM4 IO Board native schematic. J2 pin 1 corresponds to CM4 pin 101.
Unused interfaces have explicit no-connects. CM4 3.3V output supplies GPIO_VREF; internal BT is disabled.



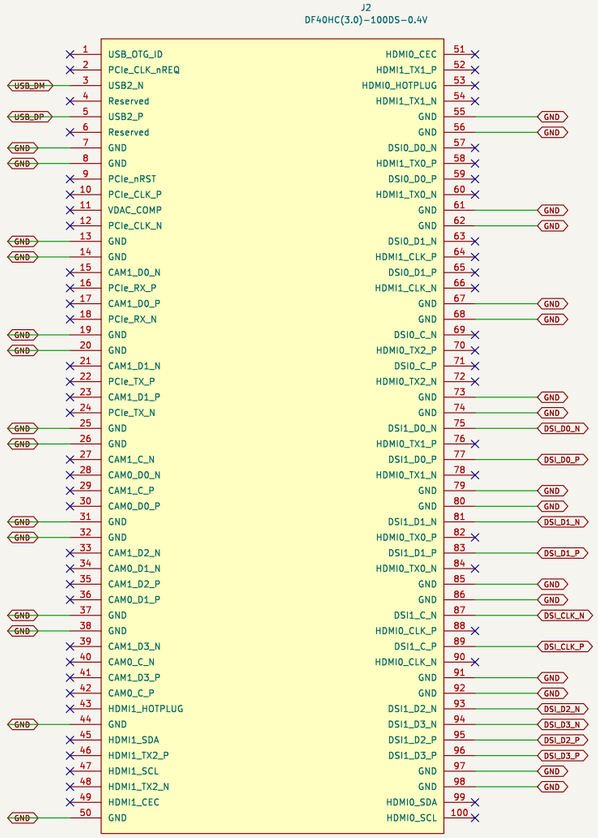
Connected schematic; hardware and manufacturing validation pending

Sheet: /CM4 GPIO and power/
File: cm4_0.kicad_sch

Title: CM4 GPIO and power

Size: A3	Date:	Rev: F
KiCad E.D.A. 9.0.7		Id: 2/8

CM4 with eMMC (not Lite): two Hirose 100-position sockets, 33.92mm centre spacing.
Pin mapping transcribed from the official CM4 IO Board native schematic. J2 pin 1 corresponds to CM4 pin 101.
Unused interfaces have explicit no-connects. CM4 3.3V output supplies GPIO_VREF; internal BT is disabled.



Connected schematic; hardware and manufacturing validation pending

Sheet: /CM4 USB and DSI/
File: cm4_1.kicad_sch

Title: CM4 USB and DSI

Size: A3	Date:	**Rev: F**
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5.1V CORE comes from power J6. CM4 returns 3.3V only while powered.
TPS2553 protects the display supply independently; output fault is read by GPIO22.
J5 mates to the supplied 3-position display power cable: 5V / unused / GND.

J3
Power J6

C1
22u 10V

C2
22u 10V

C3
100n 25V

C4
100n 25V

C5
1u 10V

U1
TPS2553DBVR

R1
22.1k 1%

R2
10k 1%

C6
1u 10V

C7
10u 10V

J5
Display supplied power cable

#FLG14
PWR_FLAG

#FLG15
PWR_FLAG

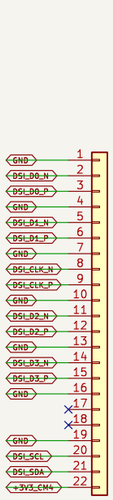
Connected schematic; hardware and manufacturing validation pending

Sheet: /Carrier power and display protection/
File: power.kicad_sch

Title: Carrier power and display protection

Size: A3	Date:	**Rev: F**
KiCad E.D.A. 9.0.7		Id: 4/8

Hirose FH12-22S-0.5SH: official CM4 IO DISP1 pinout. Use the supplied 22-to-15 DISPLAY cable.
CM4 requires dtoverlay=vc4-kms-dsi-ili9881-5inch. I2C SCL0/SDA0 are display control only.
DSI pairs stay tightly coupled over a continuous ground plane; no ground-plane splits beneath pairs.







Connected schematic; hardware and manufacturing validation pending

Sheet: /DSI1 four-lane display socket/
File: display.kicad_sch

Title: DSI1 four-lane display socket

Size: A3	Date:	**Rev: F**
KiCad E.D.A. 9.0.7		Id: 5/8

USB 2.0 peripheral to rear hub port 2. CM4 eMMC is programmed via this link with RPIBOOT grounded.
VBUS is sensing only. 22k / 33k divider keeps GPIO24 below 3.3V at 5.25V VBUS.
Never connect the PC VBUS to the CM4 5V power rail.

J4
USB IO J8



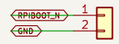
R5
22k 1%

R6
33k 1%

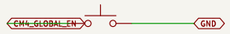
C8
10n 25V



J9
Fit jumper only for eMMC recovery



SW1
CM4 RESET



Connected schematic; hardware and manufacturing validation pending

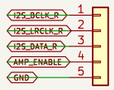
Sheet: /USB peripheral audio and recovery/
File: usb.kicad_sch

Title: USB peripheral audio and recovery

Size: A3	Date:	**Rev: F**
KiCad E.D.A. 9.0.7		Id: 6/8

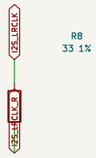
GPIO18 BCLK, GPIO19 LRCLK, GPIO21 DATA, GPIO26 amplifier enable.
J6 connects ONLY to audio J3 pins 3–7. Power is supplied separately from power J7.
Ground is carried with I2S; keep the harness short and away from switching inductors.

J6
Stereo audio J3 pins 3–7

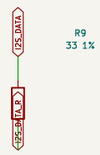




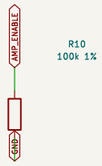
R7
33 1%



R8
33 1%



R9
33 1%



R10
100k 1%

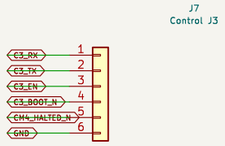
Connected schematic; hardware and manufacturing validation pending

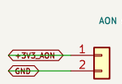
Sheet: /Stereo I2S output/
File: audio.kicad_sch

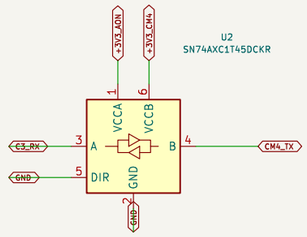
Title: Stereo I2S output

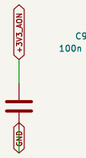
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KiCad E.D.A. 9.0.7		Id: 7/8

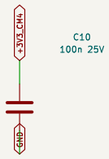
SN74AXC1T45 partial-power-down isolation: VCCA=AON, VCCB=CM4.
UART stays high-impedance when either supply is off. DIR=0 for CM4 TX to C3 RX; DIR=1 for C3 TX to CM4 RX.
Open-drain reset, boot and final-halt outputs cannot back-power CM4 GPIOs.



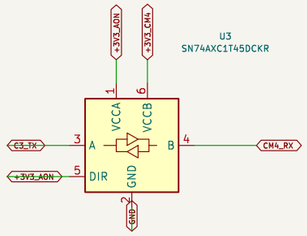


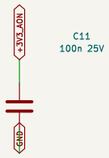


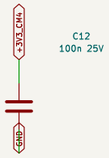




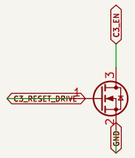


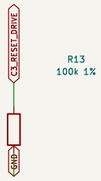


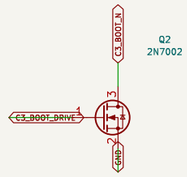


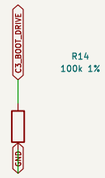


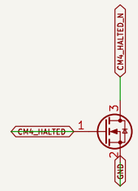


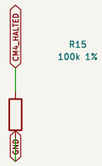














Connected schematic; hardware and manufacturing validation pending

Sheet: /Isolated CM4 to always-on bridge/
File: bridge.kicad_sch

Title: Isolated CM4 to always-on bridge

Size: A3	Date:	Rev: F
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